

Read each problem carefully. Show all work for each problem. No electronic devices allowed!

1. (16pts) Consider vectors $\vec{u} = \langle 2, -1, 2 \rangle$, $\vec{v} = \langle 1, 3, -1 \rangle$ and $\hat{i} = \langle 1, 0, 0 \rangle$. Calculate the following:

a) (5pts) $\vec{u} \cdot \vec{v}$

b) (6pts) $\hat{i} \cdot (\vec{u} \times \vec{v})$

c) (5pts) Find the **scalar component** of vector $\vec{v}$ along vector $\vec{u}$.

2. (16pts) Consider points $P(1, -1, -2)$, $Q(-2, 0, -1)$, $R(0, -2, 1)$:

a) (8pts) Find the area of the triangle determined by the points P , Q and R .

b) (8pts) Write down the equation of the plane containing these three points.

3. (8pts) Determine whether the following two lines are parallel, skew, or intersecting:

Line 1: $x = -t$, $y = 2t + 1$, $z = 3$.

Line 2: $x = s - 1$, $y = -2s$, $z = -3s$.

4. (12pts) Consider the line $\langle 1 + 4t, 2 + t, t \rangle$ and the plane $x - 2y - 2z = 1$:

a) (6pts) Determine whether the line intersects the plane.

b) (6pts) If they do intersect, find the intersection point. If they do not, find the distance between this line and this plane.

5. (12pts) Consider the planes $3x + 2y - z = 4$ and $2x - y - 3z = -2$:

a) (8pts) Find the angle between these two planes.

b) (4pts) Find any point belonging to their intersection.

6. (12pts) Consider the following quadric surface:

$$z^2 - y^2 - 4z - 2y - x^2 = 0$$

a) (8pts) Write this equation in standard form and identify this surface.

b) (4pts) Sketch the intersection of this surface with the plane $z = 4$.

7. (12pts) Consider the space-curve $\vec{r}(t) = 2t^2\hat{i} + \frac{t}{t-2}\hat{j} + (\ln t)\hat{k}$:

a) (8pts) Find the velocity vector at $t_0 = 1$.

b) (4pts) Find the equation for the line tangent to this curve at $t_0 = 1$.

8. (12pts) Compute the following expressions:

a) (8pts) $\int_0^1 [4e^{2t}\hat{i} + 3\sqrt{t}\hat{j} + \cos(\pi t)\hat{k}] dt$.

b) (4pts) $\frac{d}{dt} [(\vec{a} - t\vec{b}) \times (\vec{a} + t\vec{b})]$ where $\vec{a}$ and $\vec{b}$ are constant vectors.